



清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development, Tsinghua University

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Assessment of Iron and Steel Technologies from A Global Perspective

Tsinghua University & Saudi Aramco

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清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development, Tsinghua University

Presenter: Yunchuan Liu, PhD student

May 18, 2026

Project Timeline

	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July
Research Proposal & preparations																								
Overview of global steel trade & technology assessment																								
Data requirement & database investigation (Base on case India)																								
Model upgrade (Base on case India)																								
Case study: India																								
Case study: USA & Saudi Arabia																								
Case study: Australia & Japan																								
Case study: Germany & Russia																								
Case study: China-Australia steel trade route																								
Final report																								

Preview of today's presentation

- Review of the research framework
- Summary of India's case study

A. Research background

1. **Why we focus on the I&S industry?**
2. Why I&S industry is carbon-intensive? How to decarbonize?
3. Why we talk about India's I&S industry?

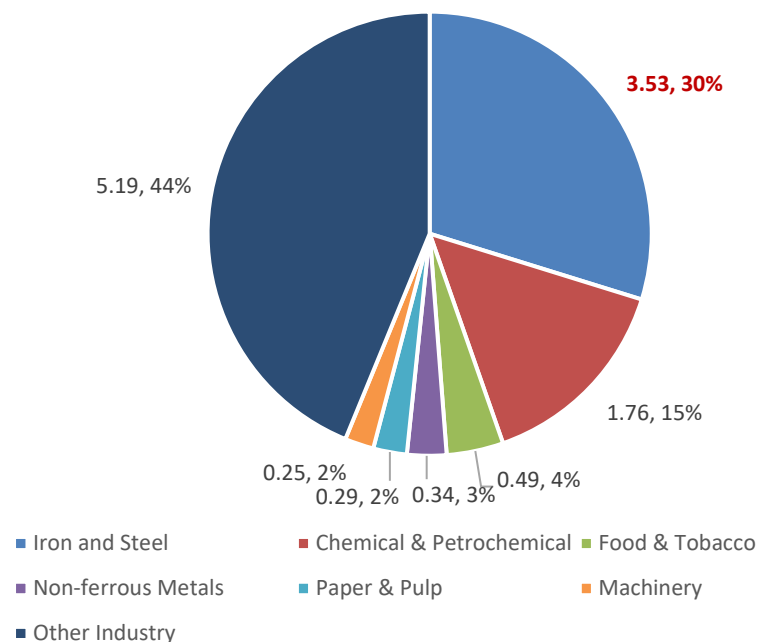
B. Research questions & steps

C. Methods & Results

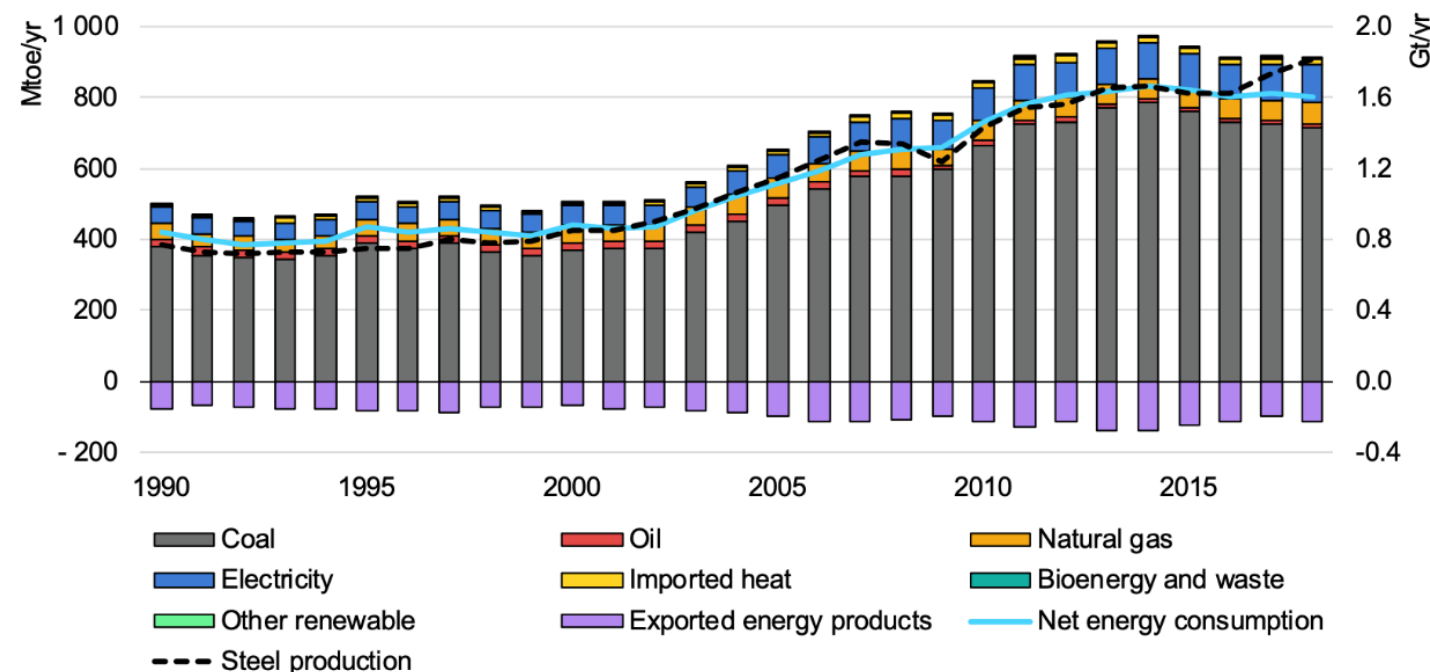
1. Characterizing H2-DRI tech series
2. National I&S transformation model
3. Scenario analysis under great uncertainty

D. Discussion & Conclusion

Why we focus on the iron & steel industry?



各工业行业排放量占比



世界钢铁行业能源消费量

- Generates ~7% of global and ~30% of industrial carbon emissions.
- Energy structure relies heavily on coal (>75%).
- Over 15 regions (excluding mainland China) have >10Mt crude steel capacity.

The I&S industry is high carbon-intensive, we need to find a greening pathway

A. Research background

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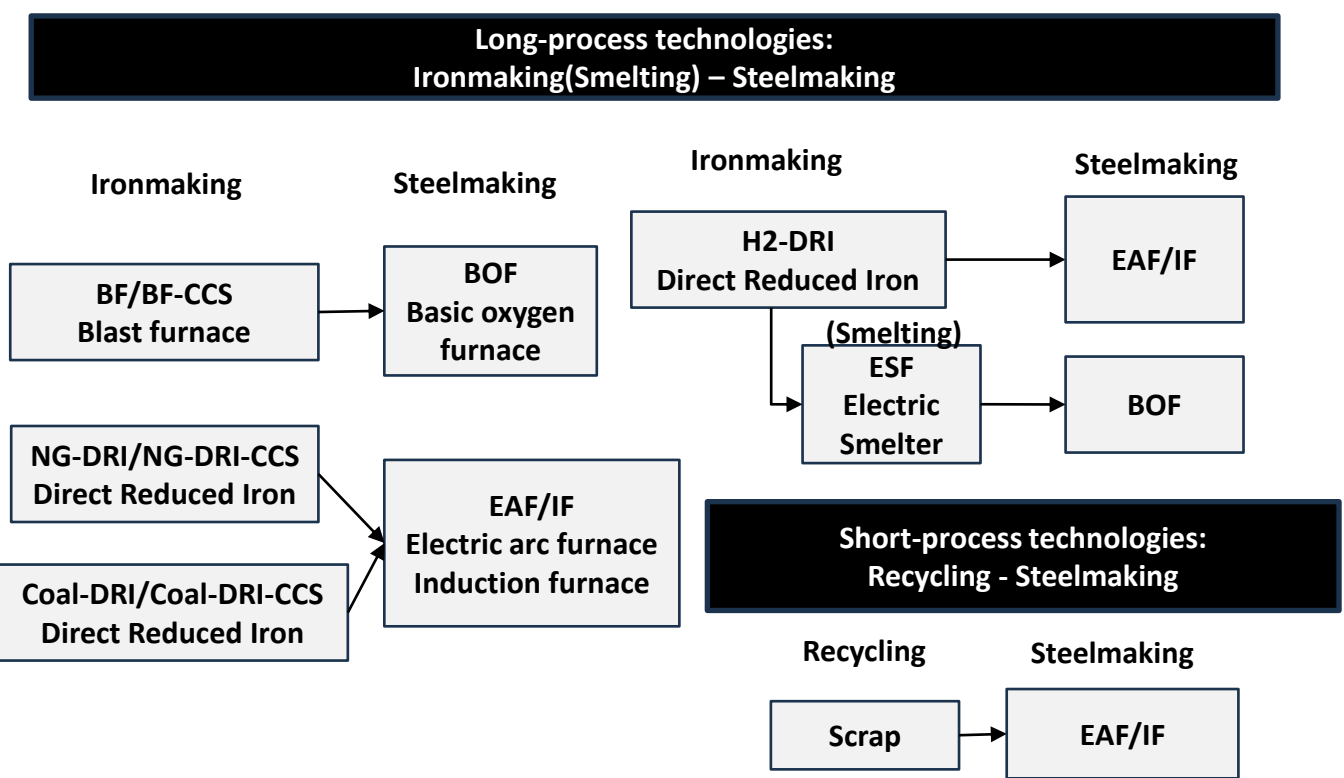
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Research background

Why I&S industry is carbon-intensive? What's the decarbonization pathway?



Tech route (without CCS)	Ore Fe%	Emission (tCO2/tcs)
Coal-DRI-EAF/IF	≥64% (BF grade)	2.7-3.1
BF-BOF	≥62% (BF grade)	2.2-2.6
NG-DRI-EAF/IF	≥67% (DR grade)	1.4-1.6
H2-DRI-ESF-BOF	≥62% (BF grade)	0.04-0.2
H2-DRI-EAF/IF	≥67% (DR grade)	0-0.10

Basic data of long-process tech routes

- **Definitions:** Box - Process; Chain – Technology
- **High-Carbon Tech:** BF-BOF and Coal-DRI drive industry emissions.
- **Low-Carbon Pathways:** Scrap-EAF(Scrap supplement limited), CCS(CO2 storage capacity limited), **Hydrogen technologies**(our focus)

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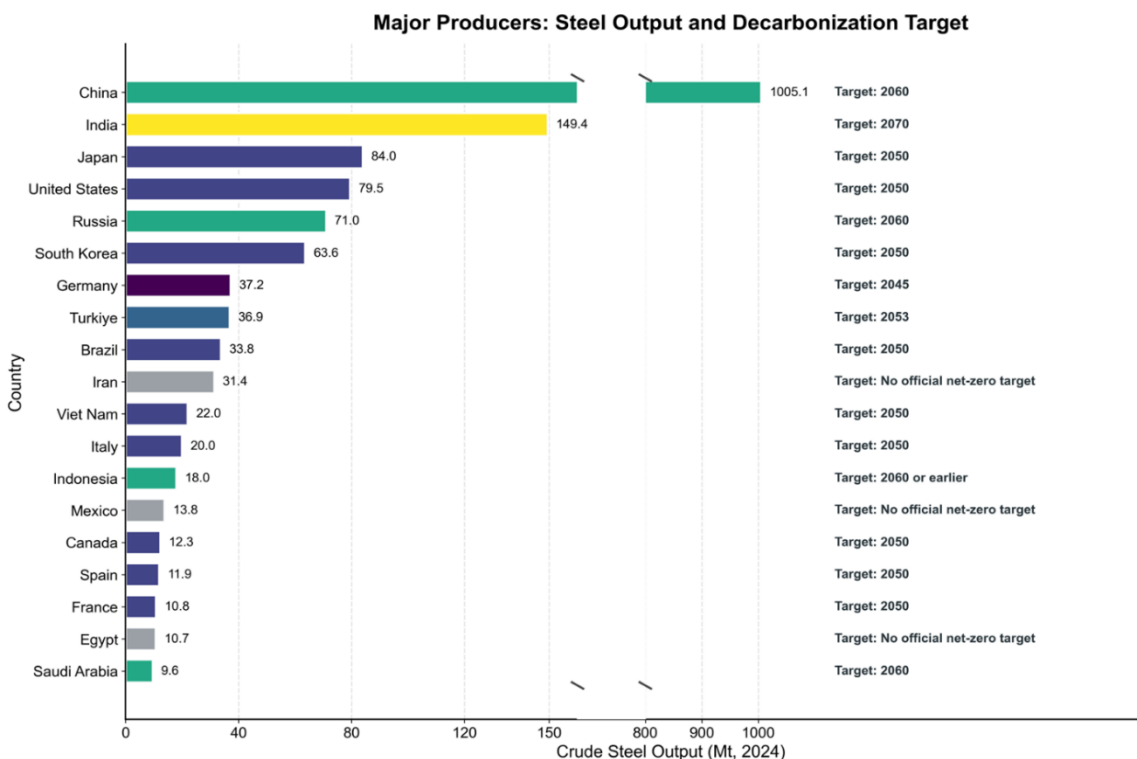
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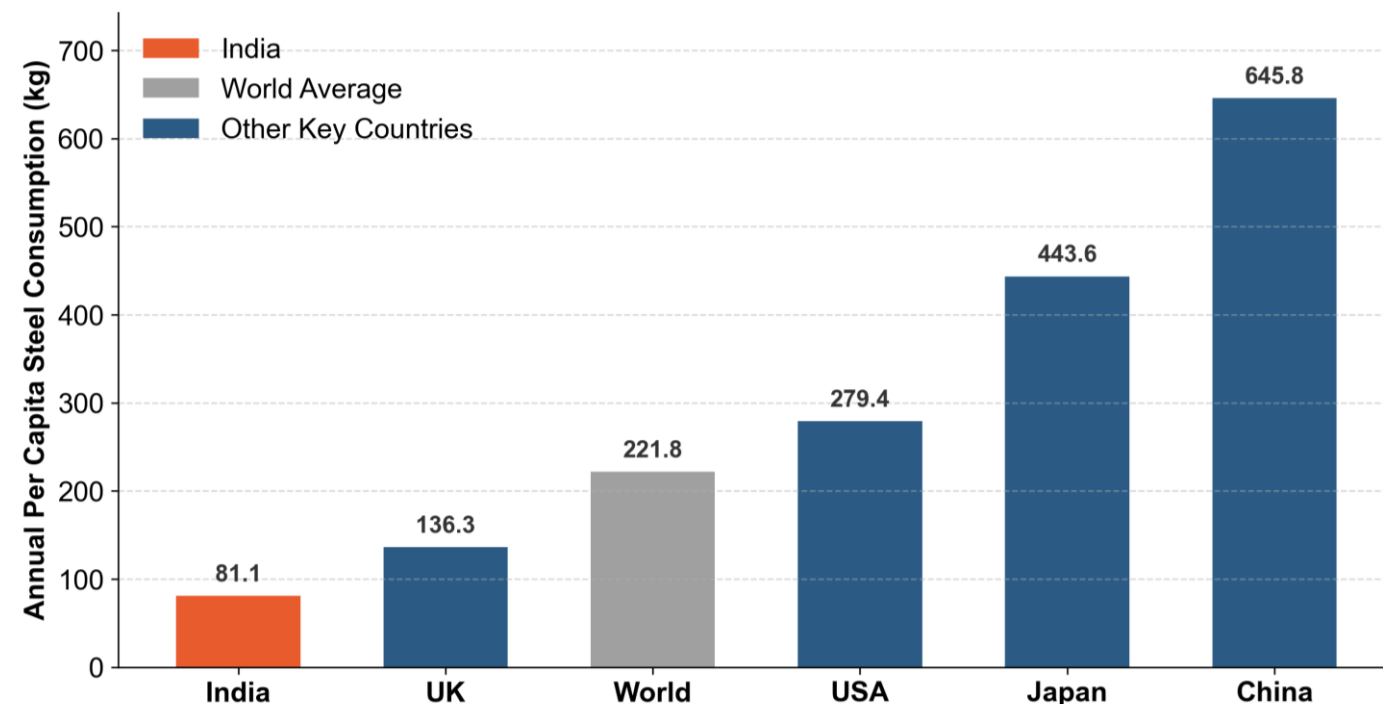
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Characteristics of India's Steel Industry: Large Scale, High Growth



Crude steel production and net-zero target years of major steel-producing countries

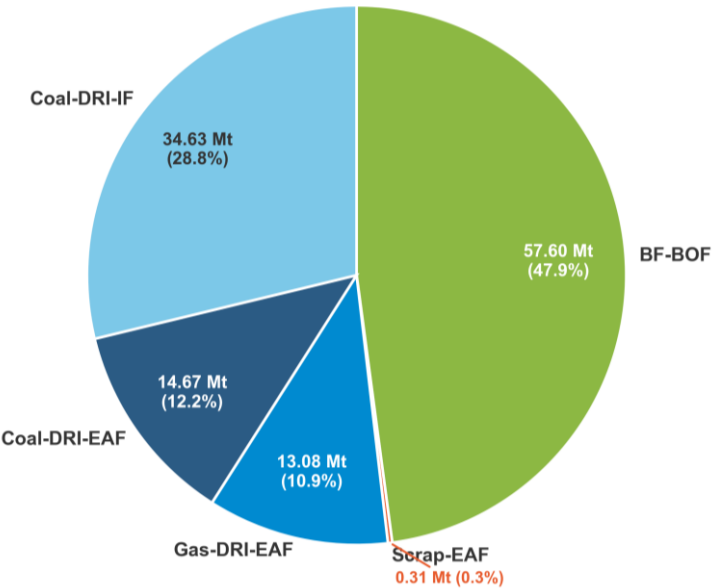


Comparison of Annual Per Capita Steel Consumption: India vs. Key Countries

- India is the **global #2 steel producer**.
- Uniquely **high growing potential** of steel demand, apart from the declining developed countries.
- Researches varies on India's **ultimate peak demand level**.

High uncertainty clouds India's future steel demand

Characteristics of India's Steel Industry: Unique techs



India's main iron & steel technical routes
(Mt crude steel production)

Technology	Coal consumption(t/tcs)	
BF (to BOF)	0.78	
Coal-DRI (to EAF)	1.0-1.3	

Process	Electricity consumption (kwh/tcs)	Fe loss (%)
EAF (100% DRI input)	450-550	3-7
IF (100% DRI input)	600-900	6-12

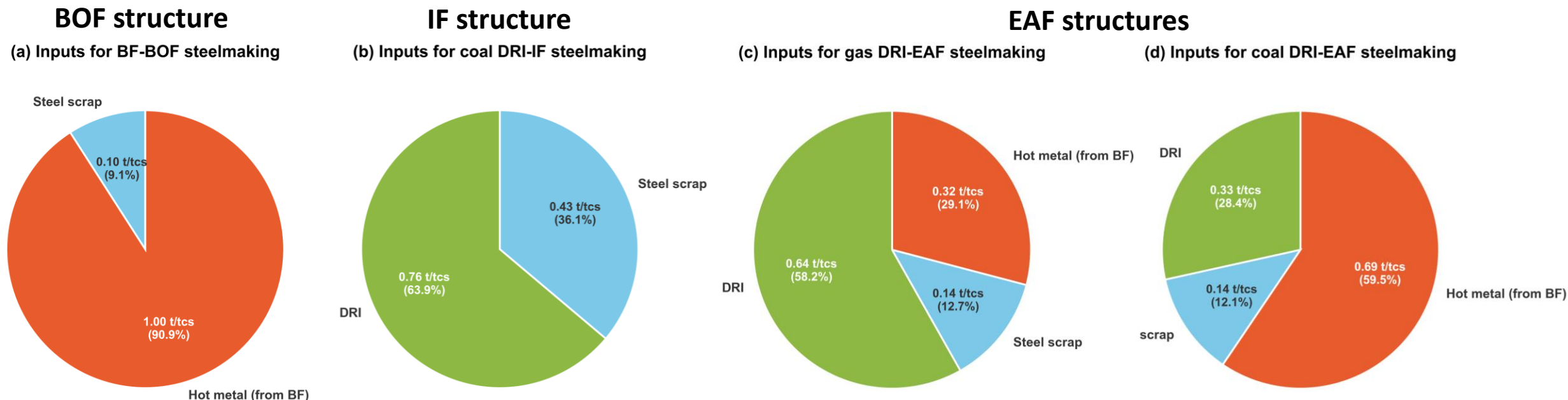
Specific efficiency metrics for selected processes of interest

Technology process	CO2 emission intensity (tCO ₂ /tcs, India)
BF-BOF	2.46
Coal-DRI-IF	2.30
Coal-DRI-EAF	2.51
NG-DRI-EAF	1.91

Typical Carbon Intensity of India's Unique Technological Pathways

- **Coal-DRI:** Higher coal use than BF-BOF. Short-term role, future phase-out.
- **IF (Induction Furnace):** Higher power use, higher iron loss than EAF. Short-term role, future phase-out.
- **Key Question: Why do Ifs still emit less carbon than EAFs in India?**

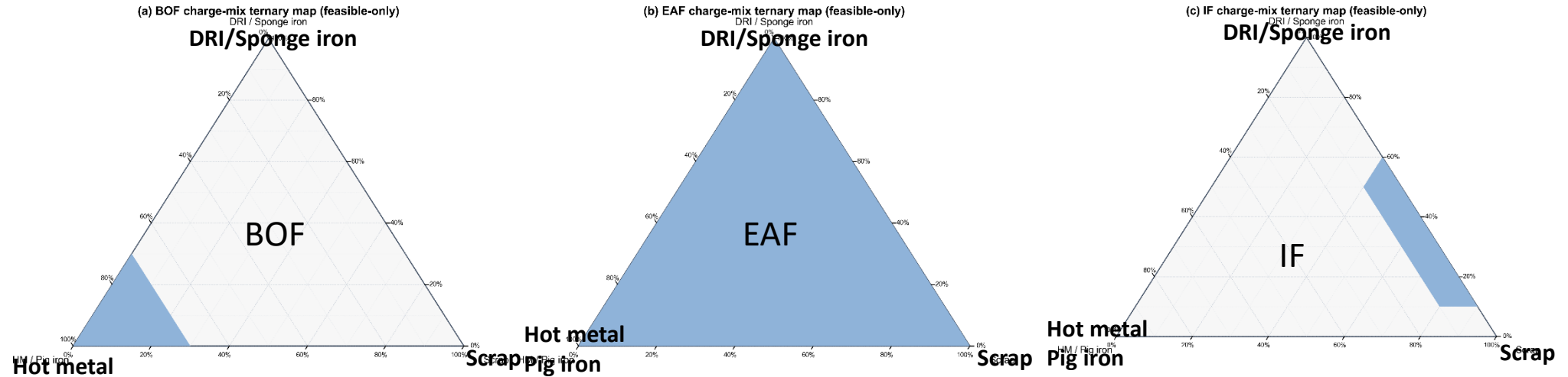
Characteristics of India's Steel Industry: Unique techs



Typical input structures of various steelmaking technologies

- **Mixture:** Mix many kinds of iron sources, and throw the mixture into the steelmaking furnace.
- **Key Insight:** Higher scrap ratio in **(b)Coal-DRI-IF** leads to lower emissions than hot metal-heavy **(d)Coal-DRI-EAF**.
- **Technical Rule:** All furnaces can accept any iron mixtures if the structure is available.

Characteristics of India's Steel Industry: Unique techs

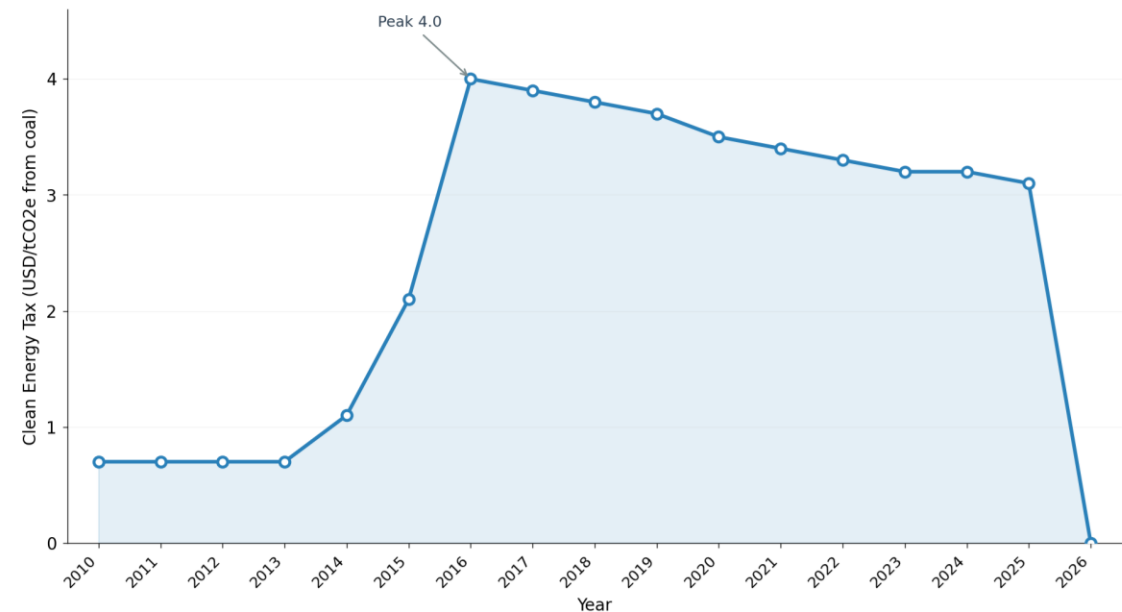


Available input structures of all steelmaking technologies in India

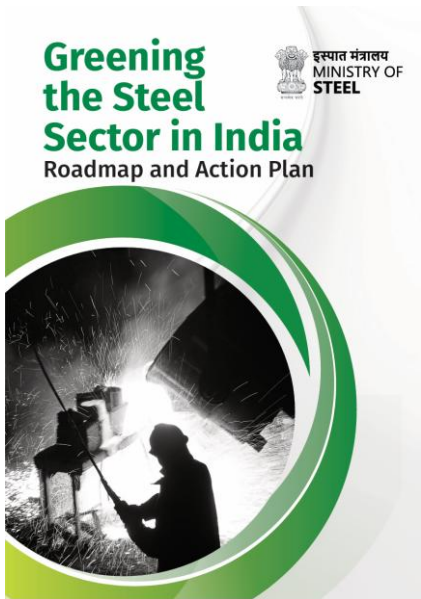
- All furnaces are flexible, so enforcing a fixed input ratio may not make sense.
- As a result, the steelmaking and ironmaking processes should be decoupled.
- Assuming this ratio constraint is universal, it can be integrated into every national study.

Research background

Characteristics of India's Steel Industry: vague carbon (pricing) policies



India's Historical Carbon Tax (USD/tCO₂ from coal)
(in the form of the Clean Environment Cess on coal)



Year	Objective
2030	45% renewable energy - Pilot plants using green hydrogen
2070	Achieve net-zero emissions for the steel industry

India's Decarbonization Goals
(Key Aspects, from the roadmap given by ministry of steel)

- The hidden carbon tax (“Clean Environment Cess” on coal) was abolished in 2026.
- The proposed CCTS(Carbon Credit Trading Scheme) remains under legislative debate.
- India aims the net-zero target of steel sector by 2070, which definitely requires an escalating carbon price.
- Few predictions exist on India's future carbon pricing.

The evolutionary pathway of carbon price is highly uncertain

India's iron and steel industry has...

Large scale, high growth

Unique techs

Massive uncertainties

Research questions: the uncertain India

1. **How** will India's steel industry achieve decarbonization?
2. How to decarbonize in countries with **booming steel demand**?
3. How to plan under **massive uncertainty** (demand, carbon price, etc.)?

Research steps

1. **Micro-Modeling**: Characterize advanced low-carbon tech (H₂-DRI tech series).
2. **Macro-Modeling**: Depict India's steel decarbonization transition.
3. **Scenario analysis**: Comprehensive scenario settings.

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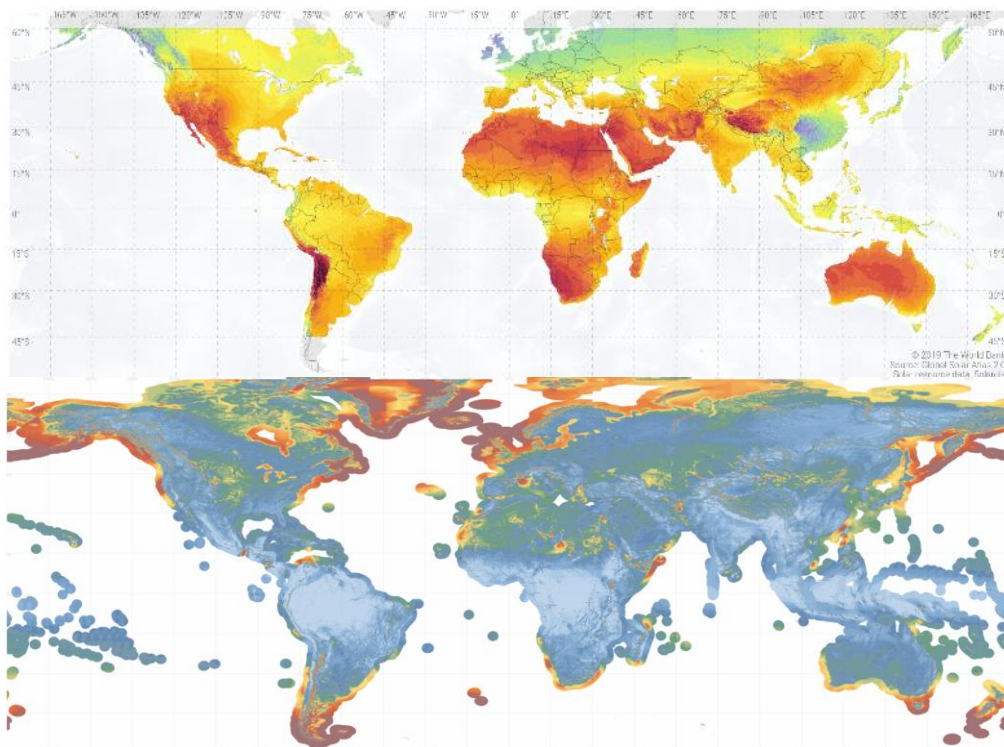
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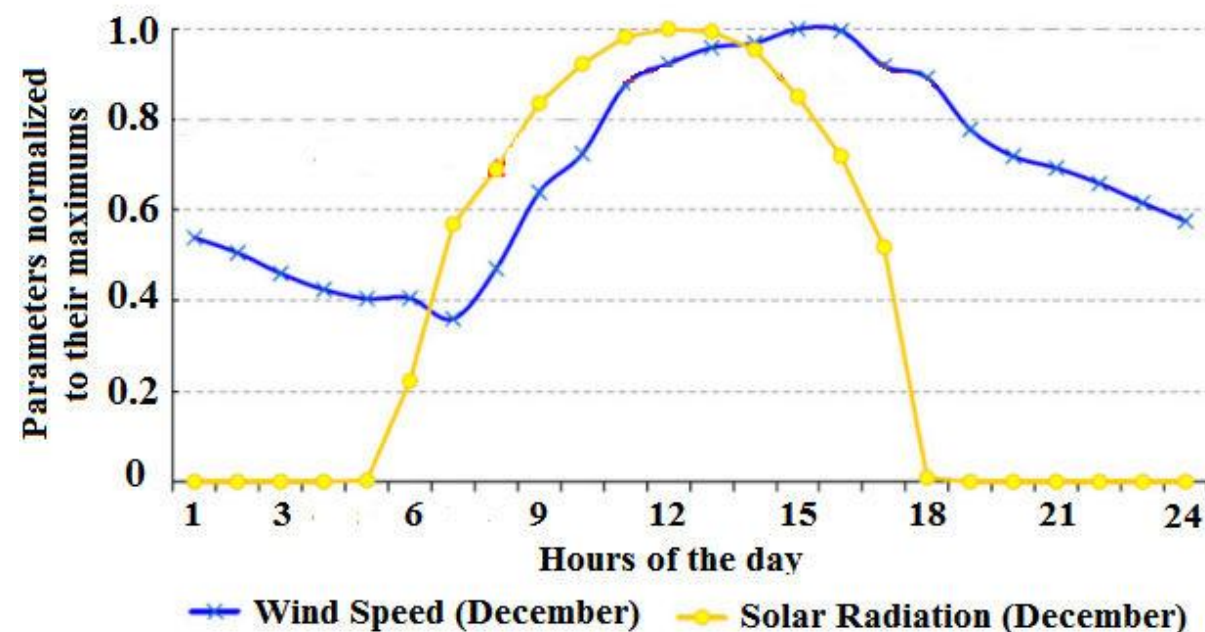
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Method 1: Characterizing H2-DRI tech series



Global Potential for PV and Wind Resources

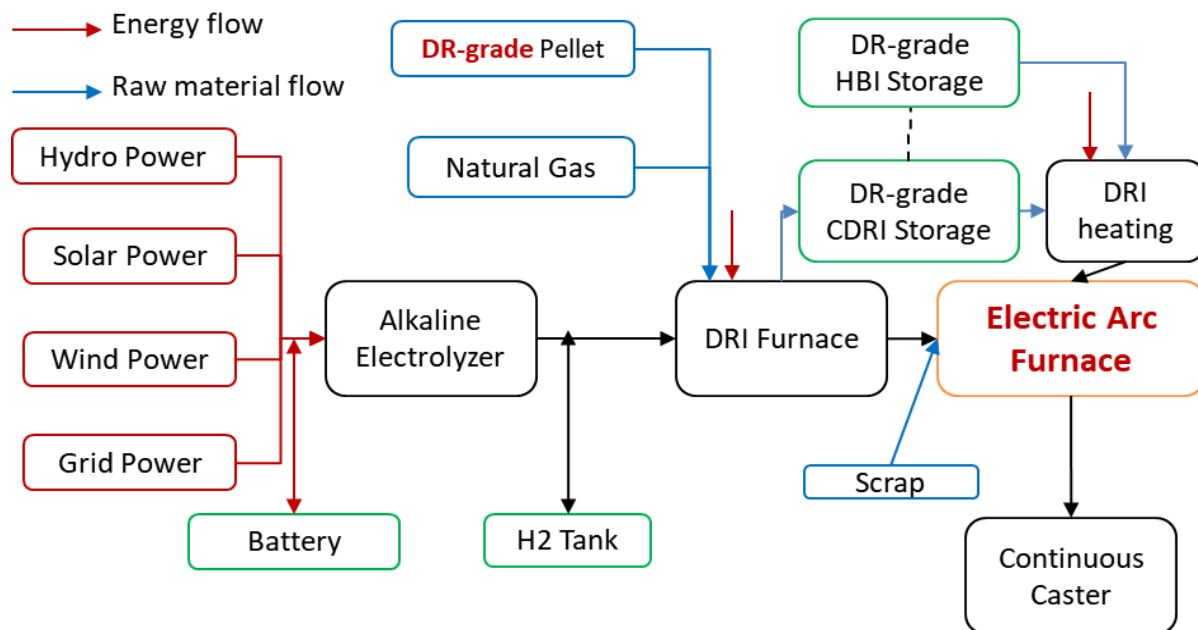


Hourly wind & solar output curves
(an example location)

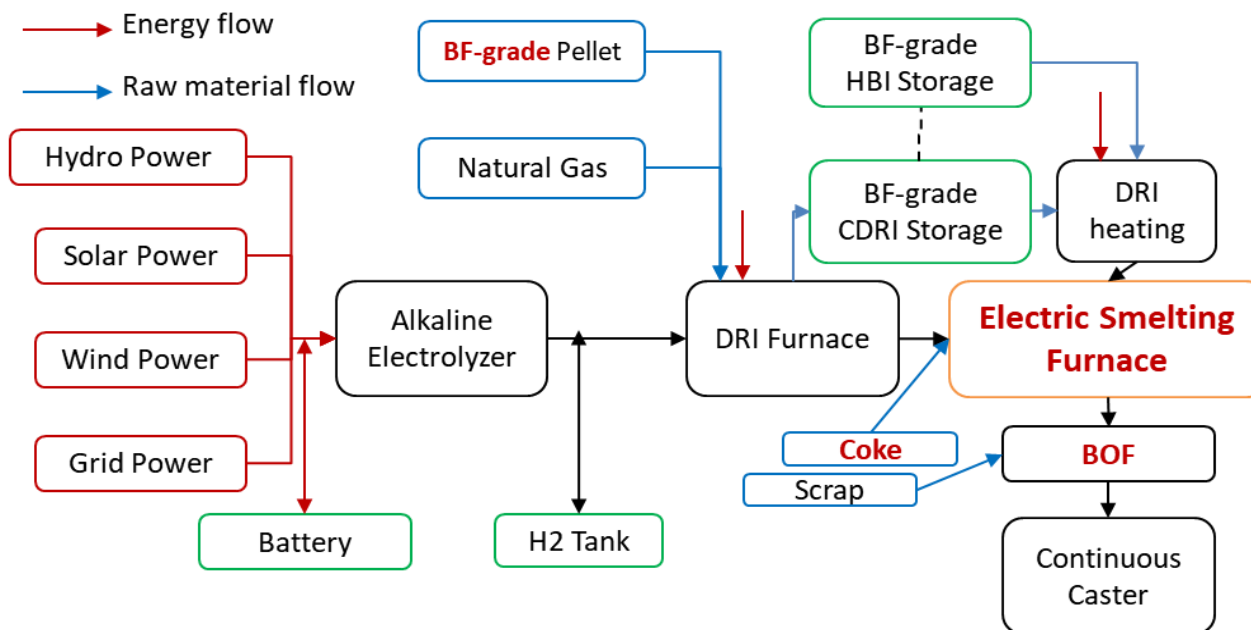
- Hydrogen metallurgy needs a high share of zero-carbon wind and solar.
- Renewable resource availability varies by locations.
- Weather-dependent and unstable renewables output requires flexibility from the industry.

Method 1: Characterizing H2-DRI tech series

Traditional H2-DRI-EAF route

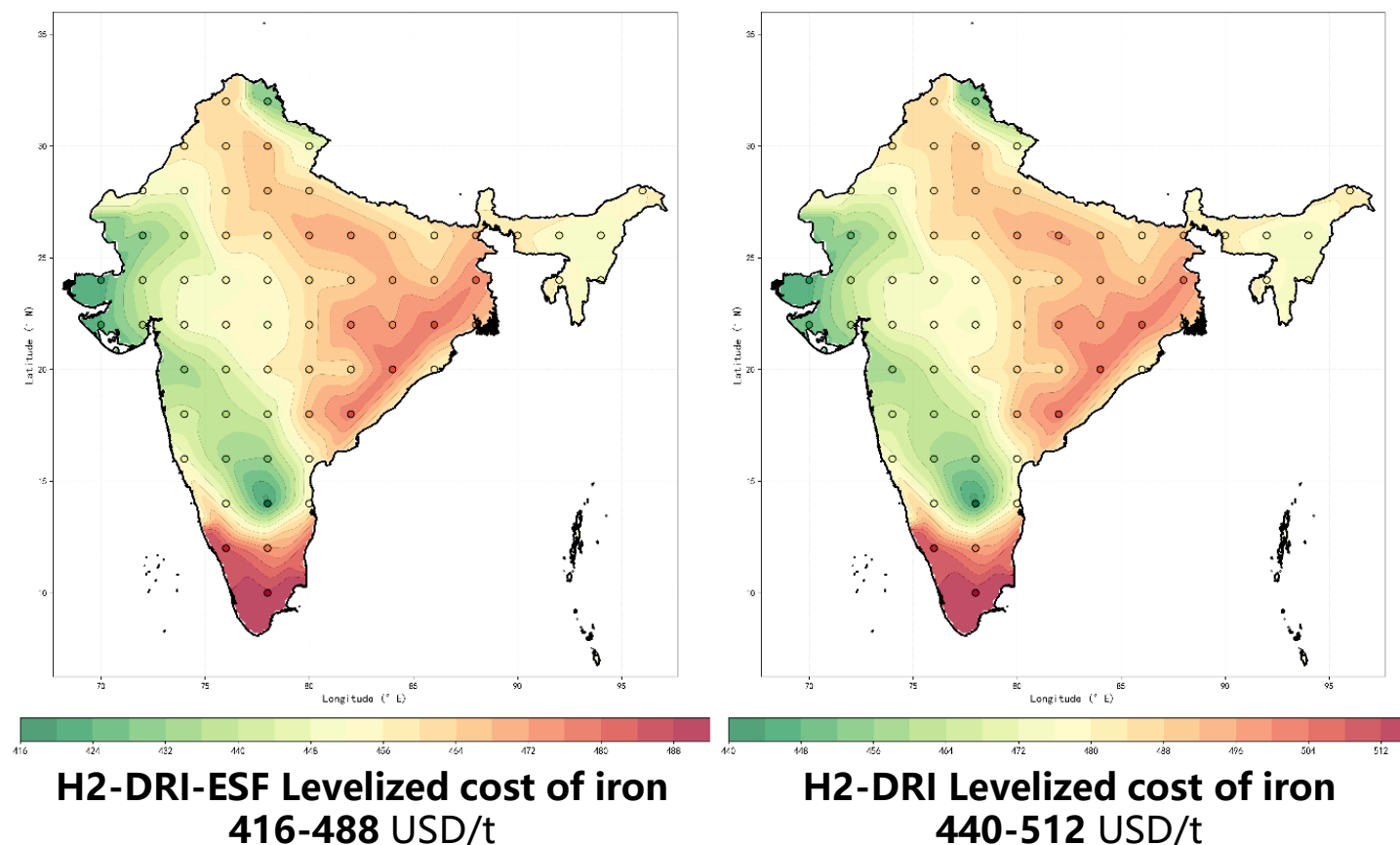


H2-DRI-ESF-BOF route



- Developed a flexible industrial process model.
- Applied the model across the entirety of India.
- Analysis combined the results with plant locations and local resource endowments.

Result 1: Characterizing H2-DRI tech series



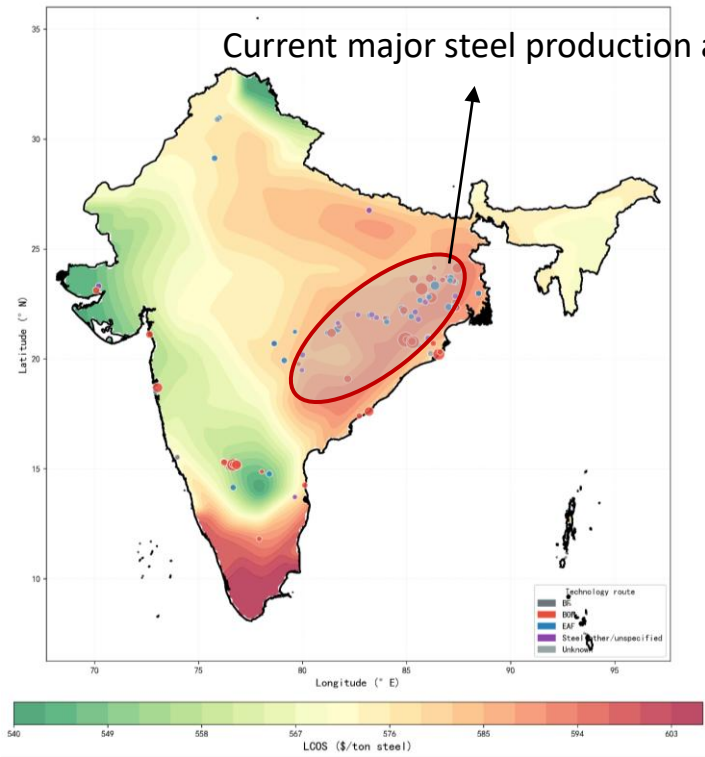
India's iron ore supply prices

Type	price	Source
import DR pellet	\$158-168	Estimates based on the Platts pellet premium
import BF pellet	\$112-118	CFR Kandla / Mundra
domestic BF pellet	\$88-98	Odisha Index

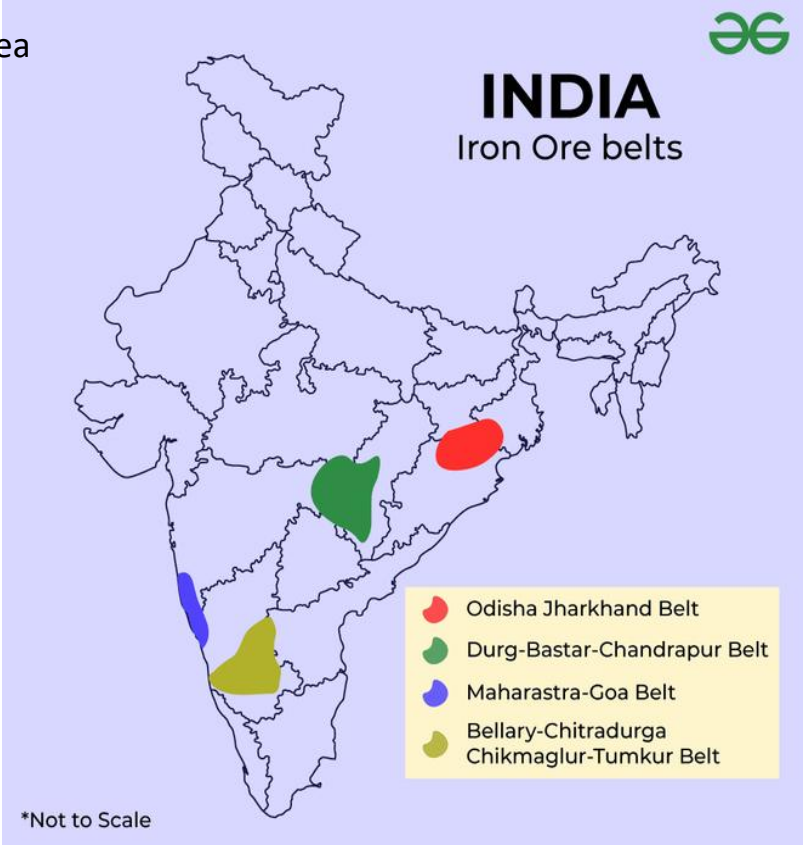
- Key metric: **Levelized cost of iron (DRI)**, which is strongly needed in other studies.
- **H2-DRI-ESF (using domestic BF pellets)** performs better over **H2-DRI (using imported DR pellets)**.
- Geography affects both costs simultaneously via renewable energy output.
- **The premium of high-grade iron ore** dictates which technology is more efficient.

Result 1: Characterizing H2-DRI tech series

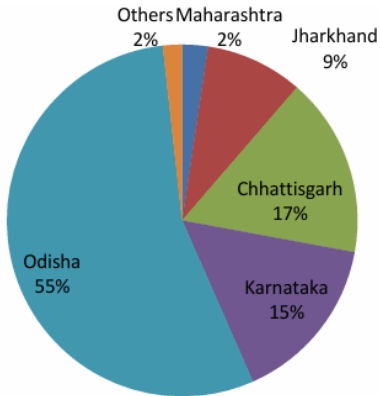
Geographical distribution: factories and ore belts



The H2-DRI-EAF LCOS heatmap (USD/tcs) with large-scale steel mills' locations (Classified by technology)



India's main iron ore belts



State-wise distribution of the Iron ore production during 2023-2024

Belt name	Fe %	Production	Domestic usage rate(%)
Odisha-Jharkhand belt	~62%*	149Mt	70~80%*
Durg-Bastar-Chandrapur belt	~65%*	46Mt	85~95%*
Magarashtra-Goa belt	~58%*	11Mt	50~70%*
Southern (Kanataka) belt	~64%*	41Mt	90~100%*

Information of India's main iron ore belts

- Plant locations mainly depend on domestic ore belt distribution and quality.
- Cost data from major production areas are used for further research.

Note: Small-scale iron ore mining areas support steel mills in other regions.

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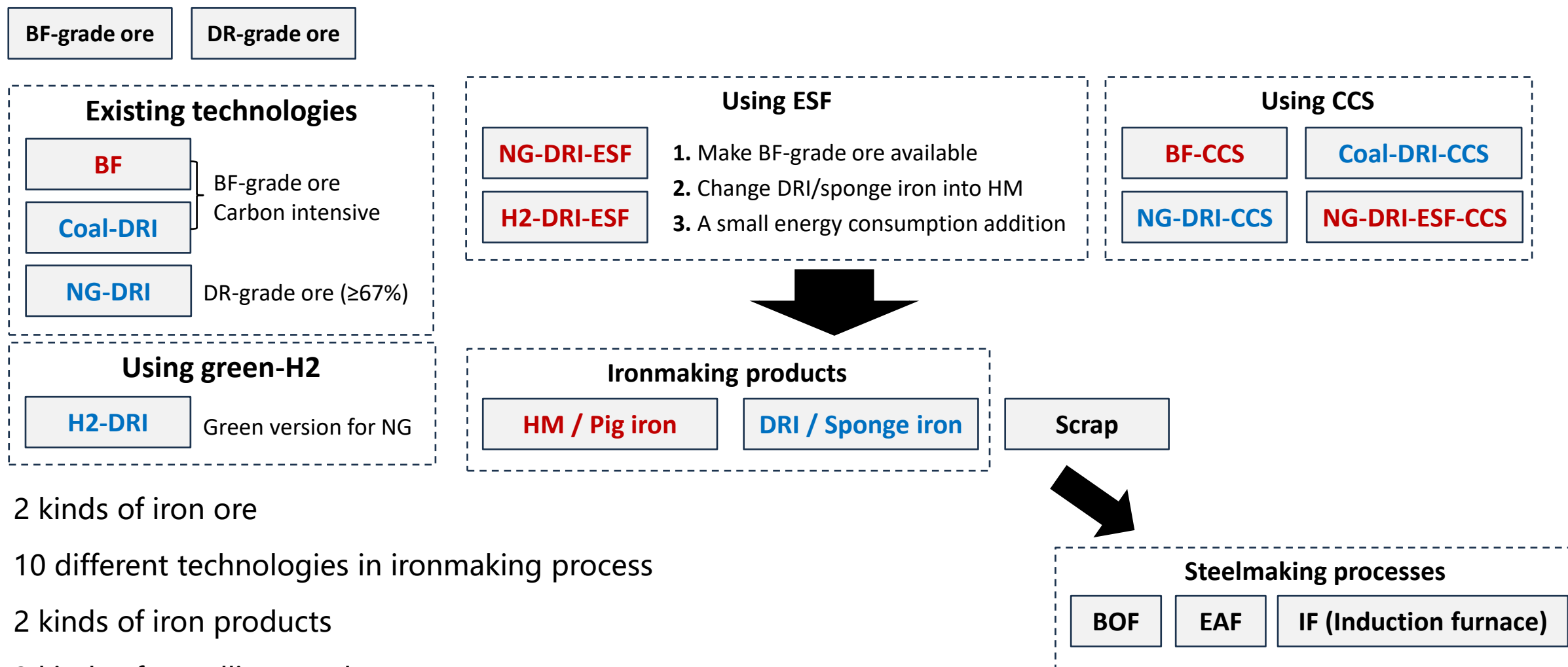
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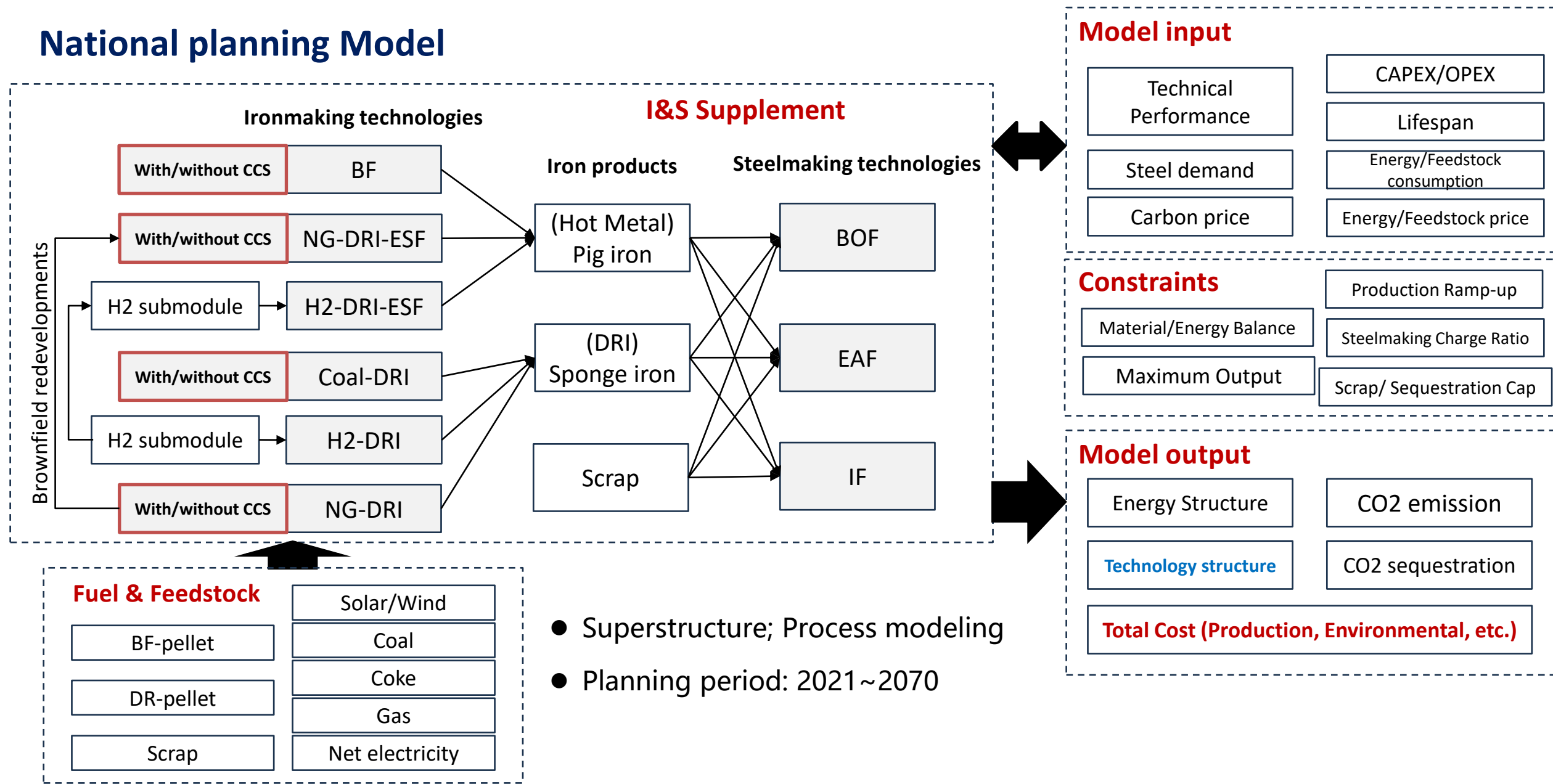
Process modeling: ironmaking & steelmaking technologies



- 2 kinds of iron ore
- 10 different technologies in ironmaking process
- 2 kinds of iron products
- 3 kinds of metallics supplement

Method 2: National I&S transformation model

National planning Model



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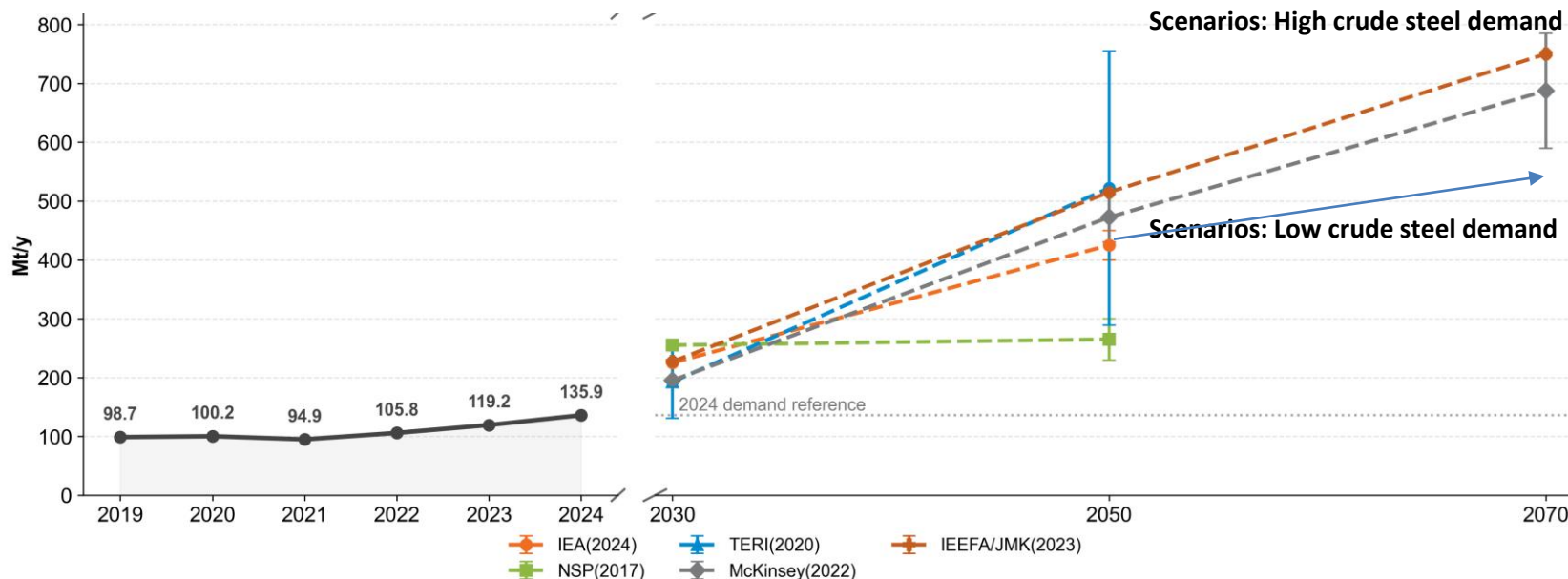
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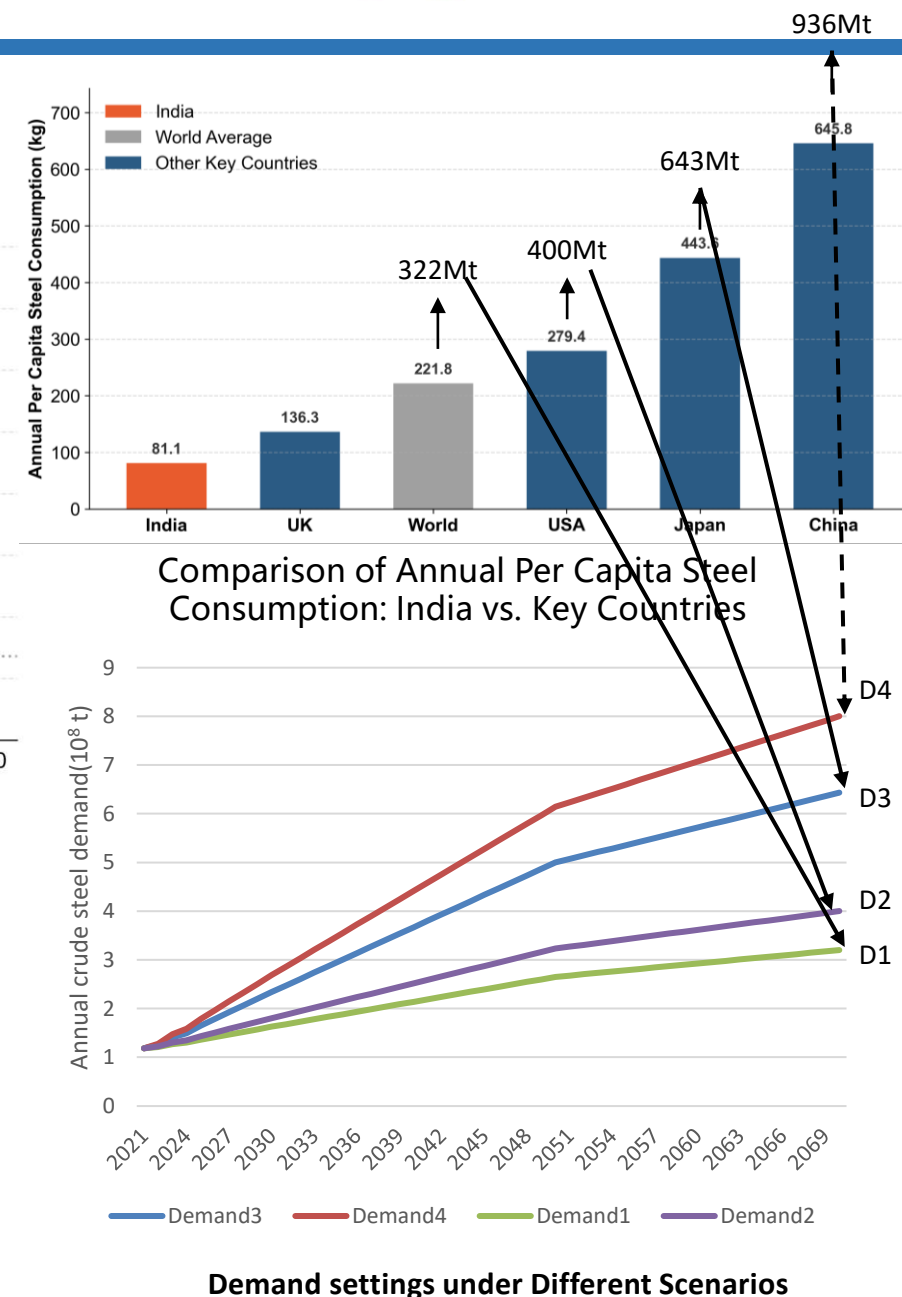
Method 3: Scenario analysis - settings

Important parameters: Steel demand



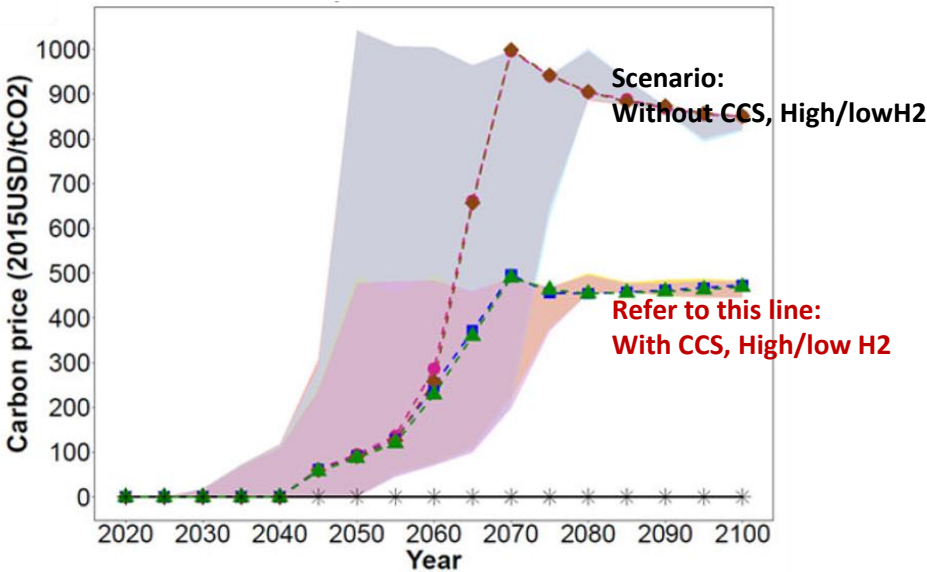
India's historical steel demand and institutional forecasts

- The majority of institutional forecasts end up to 2050.
- The **National Steel Policy (Steel ministry of India, 2017)** requires crude steel production to reach **300 Mt** in 2030 (which is almost infeasible now).
- Forecasts looking ahead to 2070 suggests India's steel demand is expected to reach a level of **600 - 800 Mt**.
- We set 4 levels of steel demand **base on the per-capita steel consumption**.



Method 3: Scenario analysis - settings

Important parameters: Carbon price



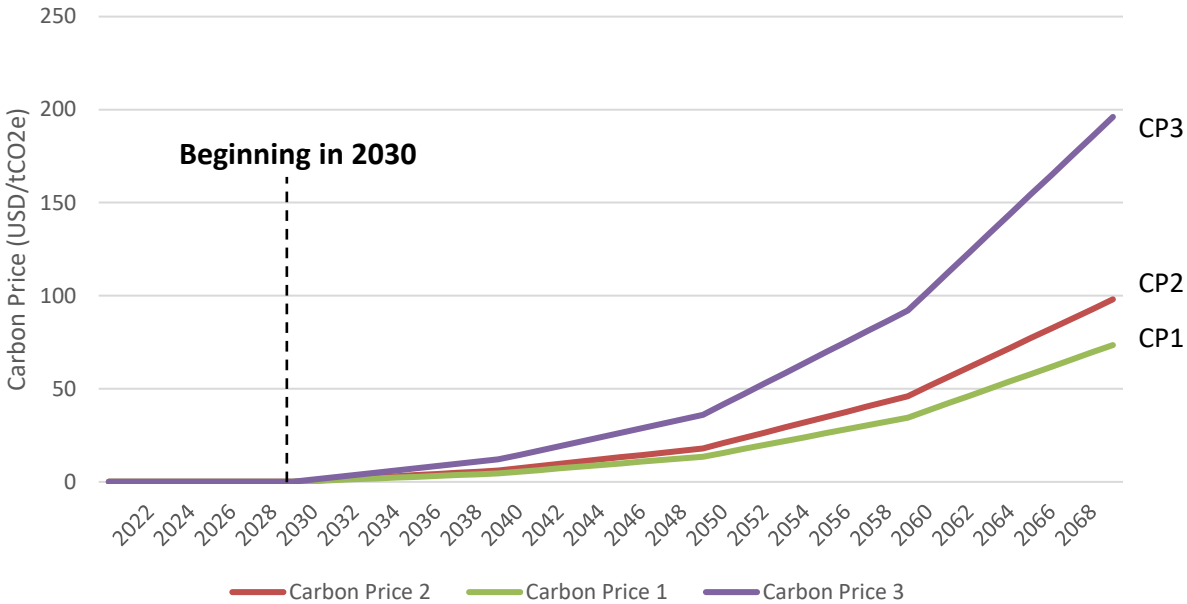
Ref.1 Carbon price across scenarios (CEEW)

- Current analyses of carbon pricing in India are really limited.
- Ref.1 is a research which focus on carbon price, but points to an extremely high number. Ref.2 present a relatively acceptable assumption.
- We build a mild **baseline levels** with lower CP according to ref.2, and set a **higher CP level** acting closer to ref.1.

TABLE 1 | Key policies in LTD scenario Cont.

POLICY		REFERENCE SCENARIO (2050)	LTD SCENARIO (2050)
CARBON PRICES USED FOR ANALYSIS			
Carbon Tax (lower bound)	Per tonne of CO ₂ in electricity and industry sector	0	\$4 to \$50 by 2050
Carbon Tax (Upper bound)	Per tonne of CO ₂ in electricity and industry sector	0	\$4 to \$100 by 2050

Ref.2 Carbon price settings across scenarios (WEA India)



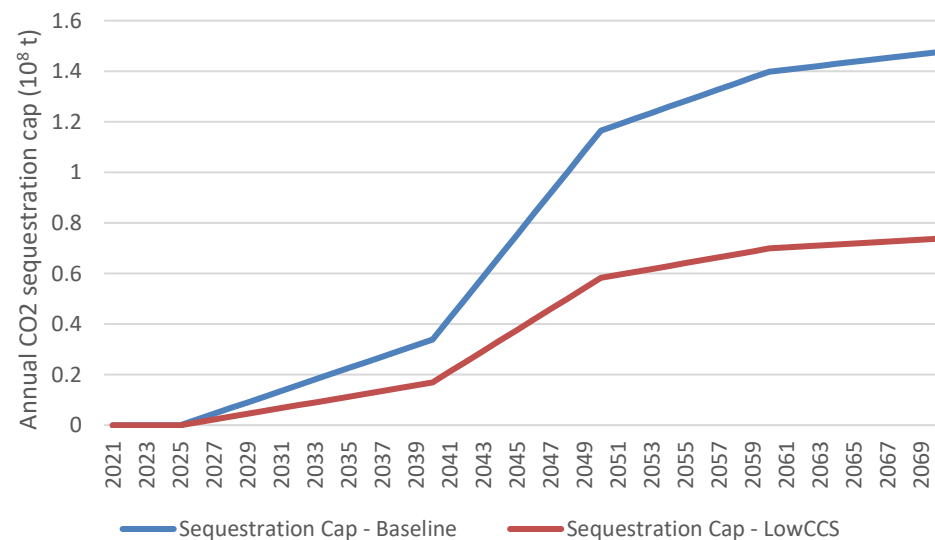
Carbon price settings under different scenarios (USD/tCO2e)

Source:

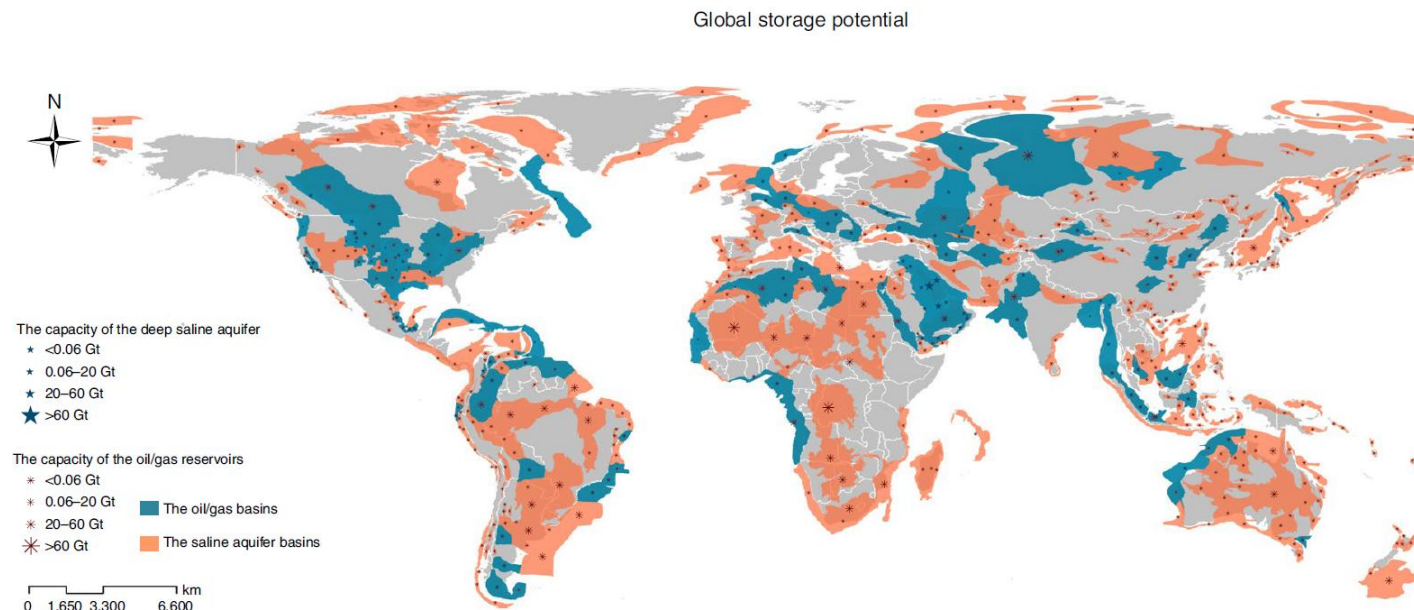
Ref1.CEEW(Council on Energy, Environment and Water). Implications of a Net-Zero Target for India's Sectoral Energy Transitions and Climate Policy. doi.org/10.1093/oxfclm/kgac001

Ref2. WEA India. Pricing carbon: Risks and opportunities for India – an analysis using India energy policy simulator

Important parameters: CO2 sequestration limit



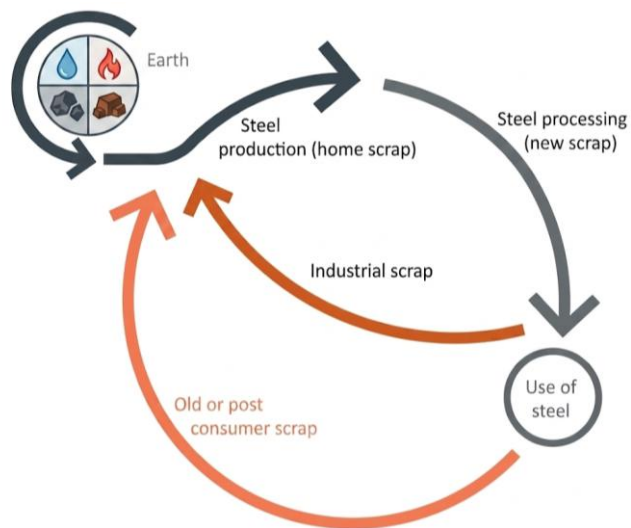
Parameter: CO2 sequestration cap



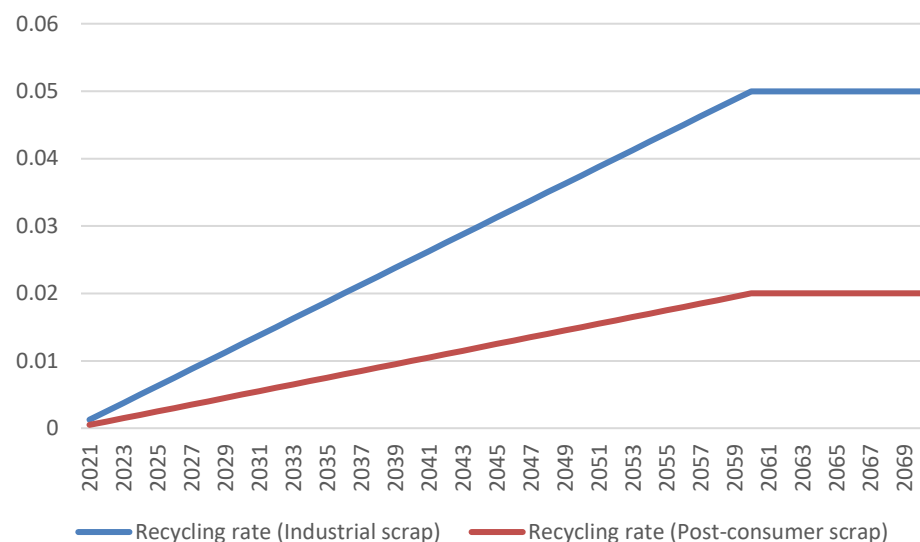
Ref. Carbon price across scenarios (CEEW)

- India's CO2 sequestration relies heavily on sedimentary basins for aquifer storage.
- India has **3.8 GtCO2 sequestration limit** according to the reference.
- **Baseline level:** The upper limit curve for carbon sequestration in India's steel industry, estimated based on reference values, assumes that the steel industry accounts for 50% of the total sequestration.
- **LowCCS level:** Assumes that the iron & steel industry accounts for 25% of the total sequestration

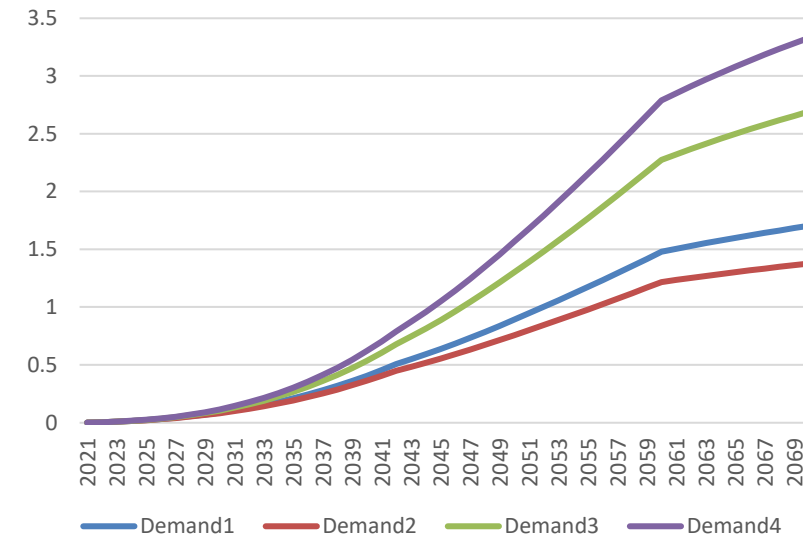
Important parameters: Scrap & recycling cap



The origins of two types of Scrap



Scrap recycling rate by types



Scrap cap under scenarios

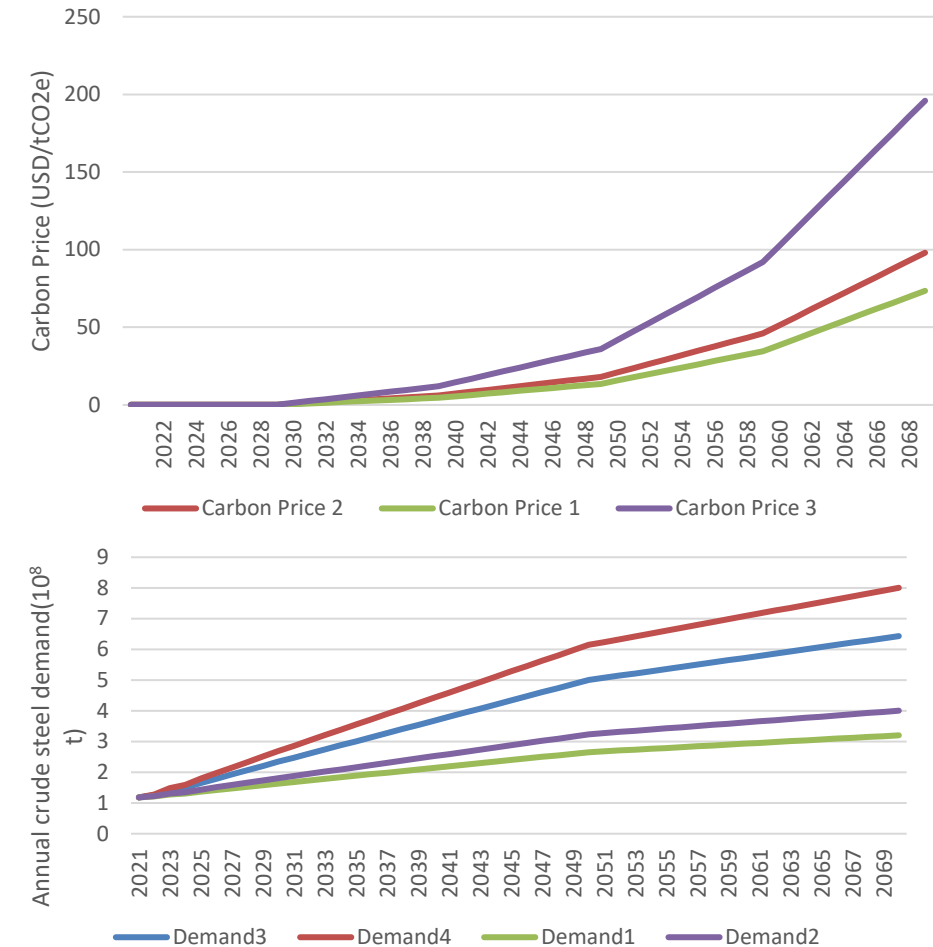
- **Industrial scrap:** Scrap recovered in a certain rate(5% in China) due to process losses during steel production.
- **Post consumer scrap:** Scrap recovered annually in a certain rate(2% in China) of the total steel inventory in society.
- According to sources, we can approximate that India initially produced low scrap in 2020, and gradually established a comprehensive scrap steel recycling system by 2060.
- Under different demand settings, the **cap of domestic scrap** supplement also varies.

Method 3: Scenario analysis - settings

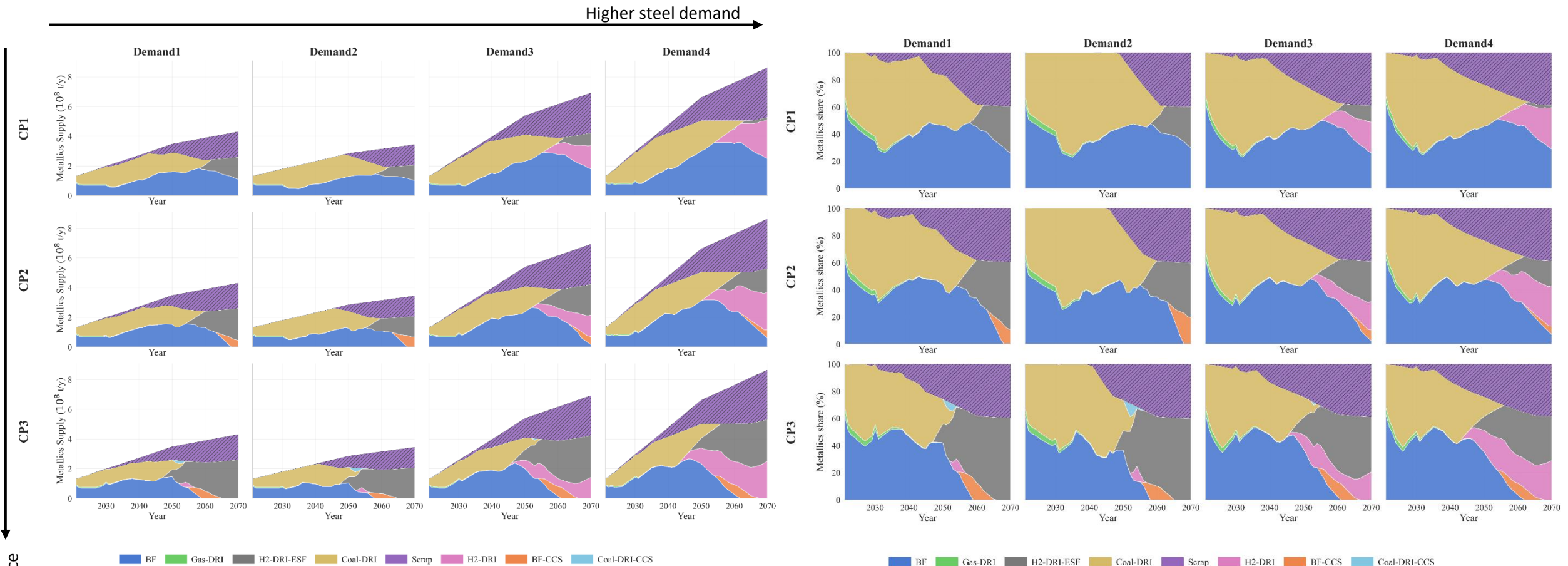
Parameter settings: conclusion

Parameter	Settings / level			
Carbon price	CP1	CP2		CP3
Steel demand	Demand1	Demand2	Demand3	Demand4
CO2 sequestration limit	Baseline		LowCCS	
others	Baseline			

- **Focus:** Carbon price & steel demand (CO2 sequestration baseline)
- **Total:** 12 Scenarios



Technology structure: Metalics supply of steelmaking process (Iron and scrap)

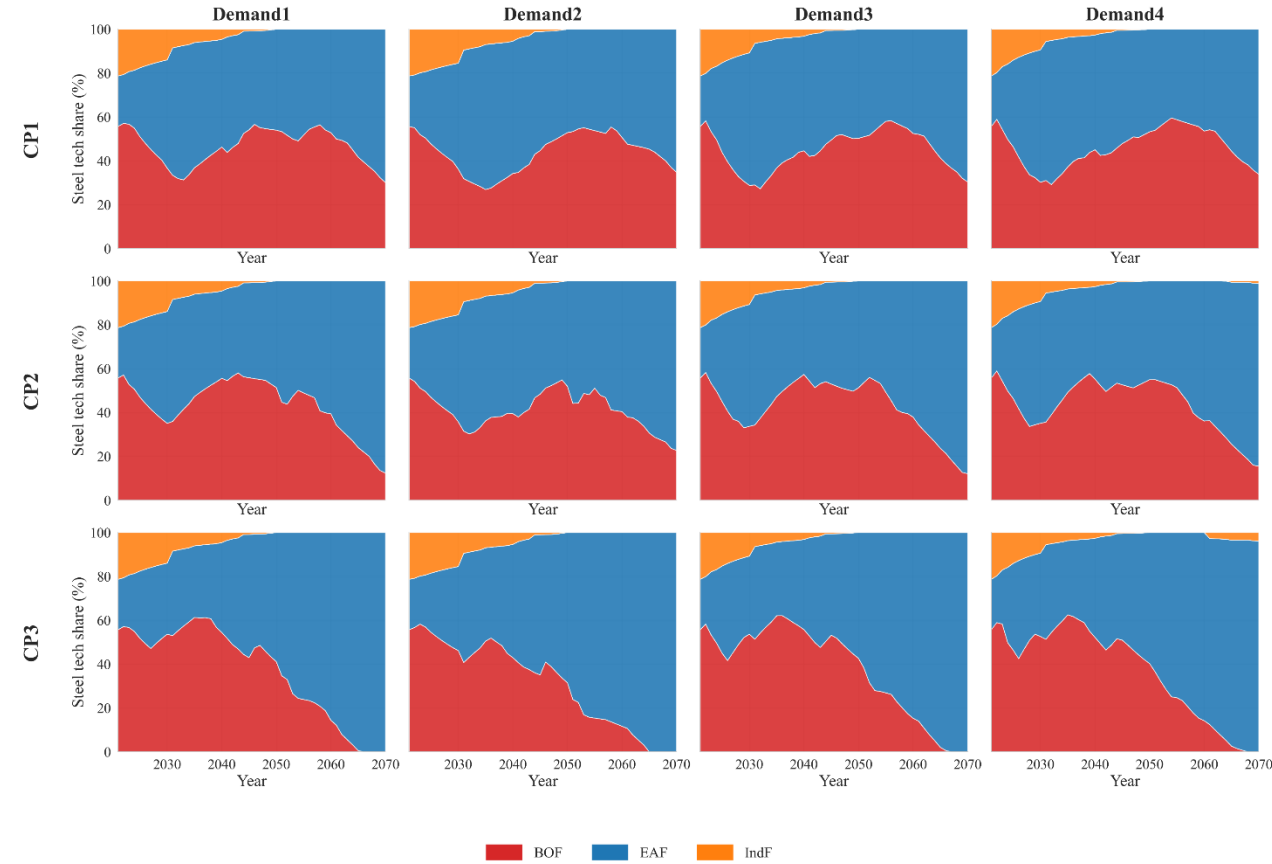
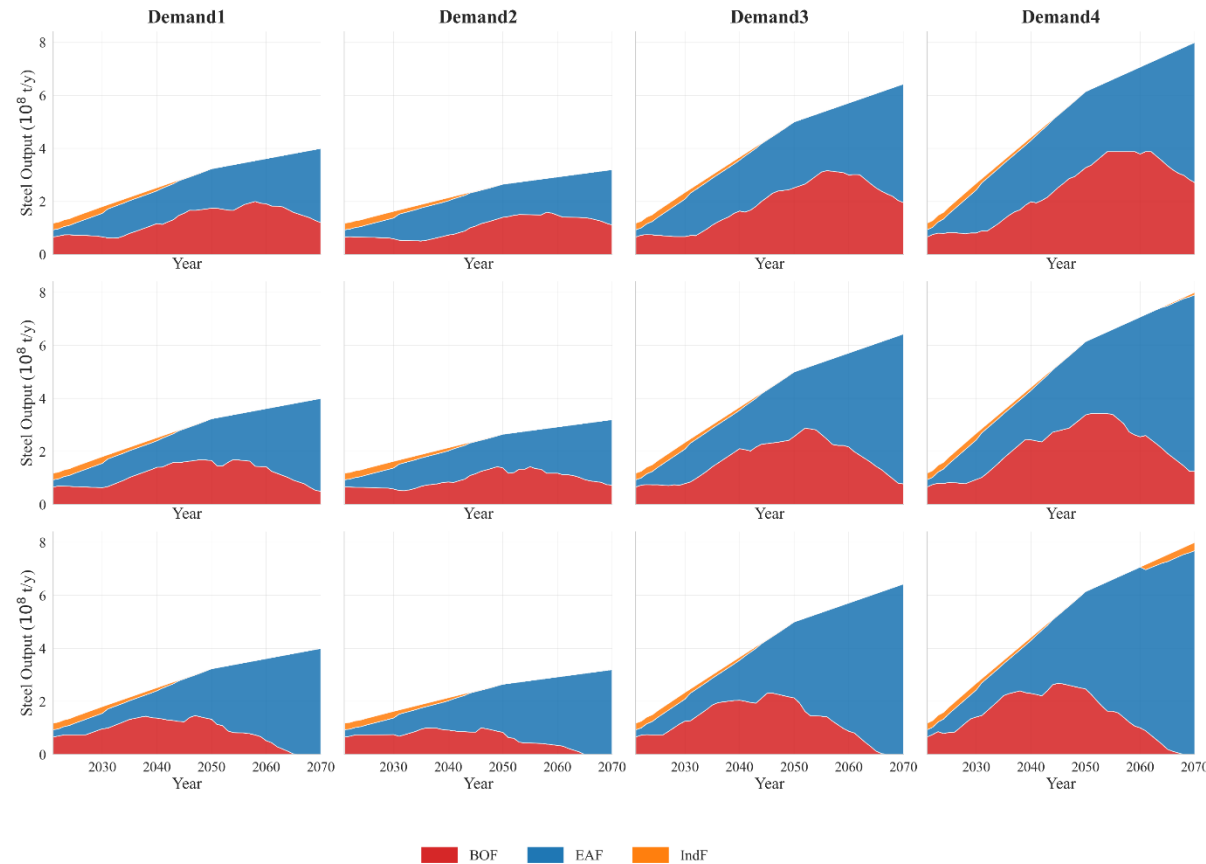


- Technology transformation: BF, Coal-DRI → H2, CCS, Scrap
- Higher carbon price (but not apparently less demand growth) leads to a quicker technology transformation.
- With the lack of cheap domestic BF-level iron ore, H2-DRI exists in the higher demand scenarios.

Technology structure: Steelmaking

Higher steel demand →

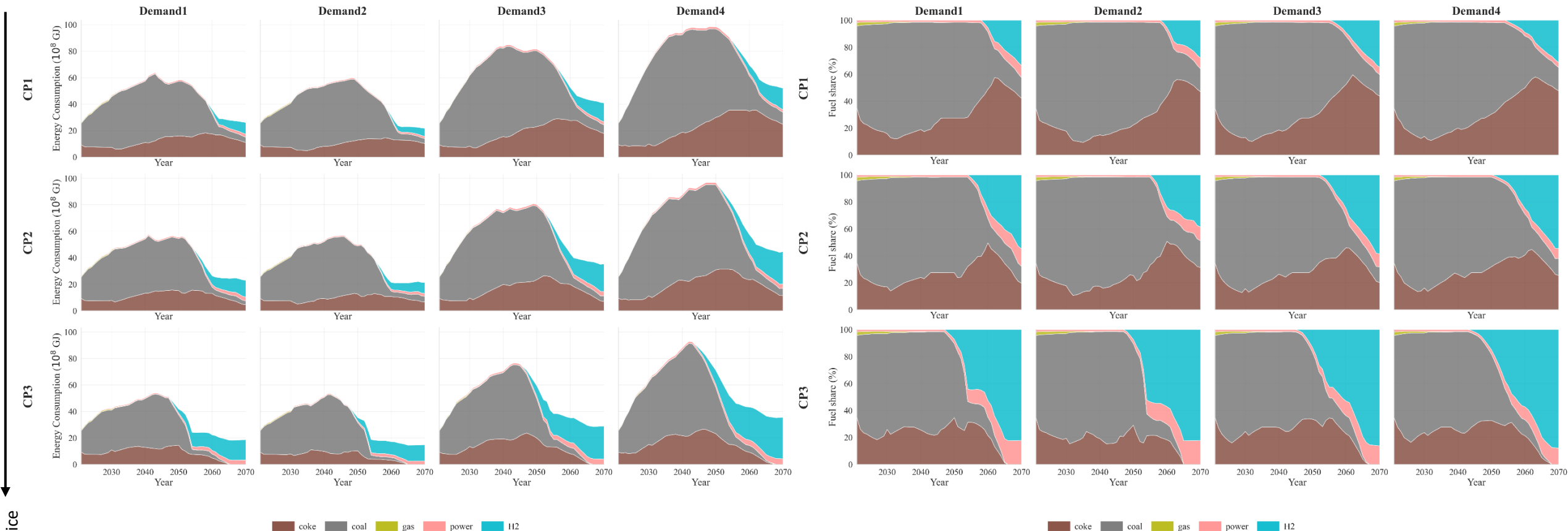
Higher carbon price



- Technology transformation: BOF,IF -> EAF
- Higher carbon price (but not apparently less demand growth) leads to a quicker technology transformation.

Energy consumption

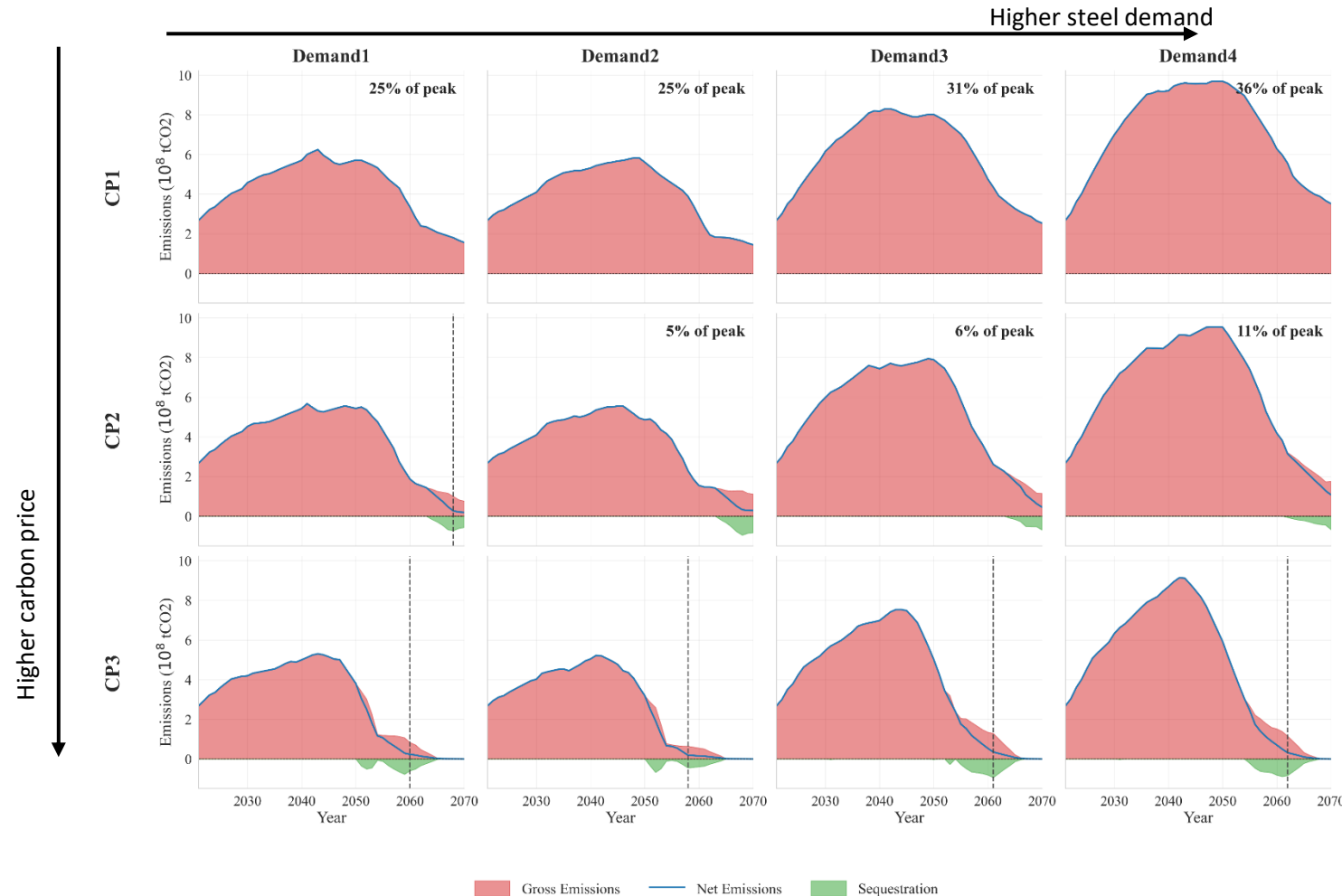
Higher steel demand →



- Energy transformation: coal, coke → power, H2
- Higher carbon price (but not apparently less demand growth) leads to a quicker energy transformation.

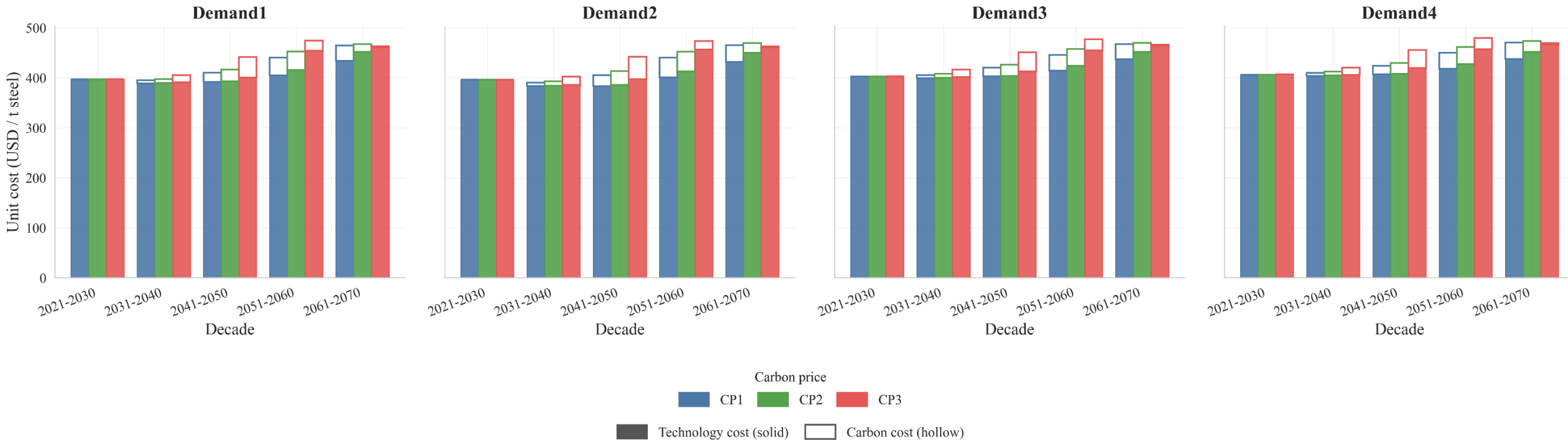
Higher carbon price ↓

Iron input structure for steelmaking technologies



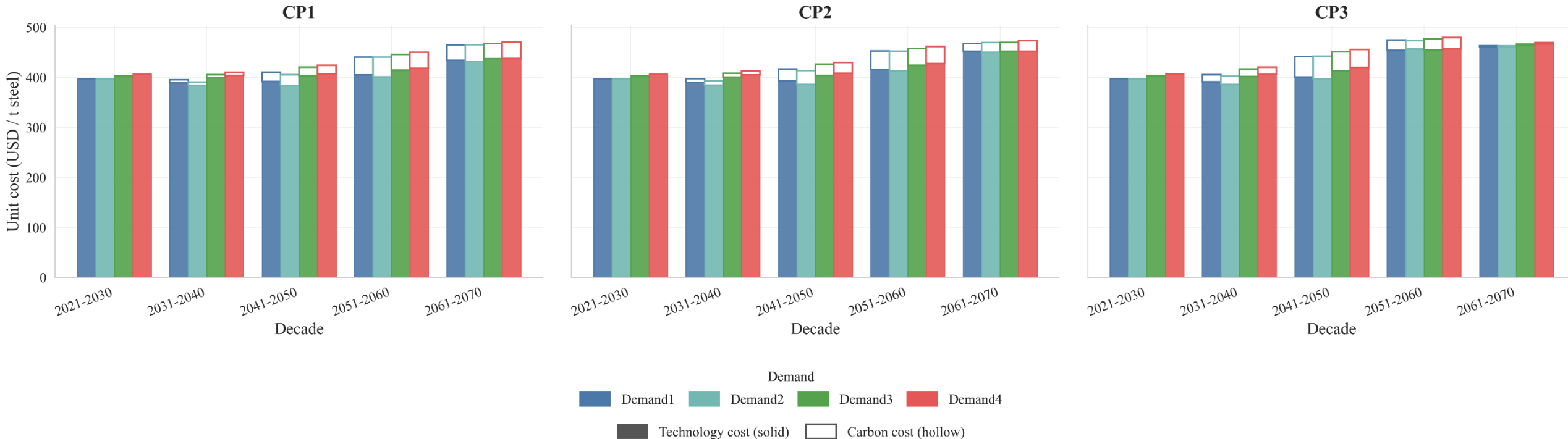
- We believe a **95% CO₂e reduction from peak** means that the **decarbonization target** has been achieved.
- **Higher cost-effectiveness and reduced demand for steel** can have a profound impact, making decarbonization more feasible and potentially achievable sooner.
- CP1 scenarios: A small CP make no sense
- CP3 scenarios: A high CP significantly drive the decarbonization
- CP2 scenarios: 1 of 4 scenarios finish the decarbonization

Total cost (Unit steel production)



- **Raising carbon price is the way to net-zero, but increase the unit cost during the transition period, especially 2041-2060 (CP2-CP3, 5% increase).**
- When carbon price increases to a certain level, since most of them have eventually completed decarbonization the technical structure is basically similar, **the unit cost in 2070 will tend to be consistent.**

Total cost (Unit steel production)



- **Raising demand growth** causes technology costs to initially decrease and then increase, and carbon costs to initially increase and then decrease.
- The absolute magnitude of these changes does not result in a significant impact.

● India is unique

- Distinct tech
- Booming demand
- High uncertainty (demand & carbon price)

● Demand Impact

- Lowering demand growth speeds up decarbonization with stable unit costs
- But a low demand growth may hurt national development.

● Carbon price Impact

- Low carbon price is ineffective for decarbonization.
- High carbon price helps achieving net-zero, but significantly increases transition costs.

● India case

- Finalizing adjustments based on feedback
- Begin writing the paper for this part of the research

● New cases

- Investigating other countries (upnext USA as planned)